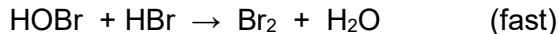
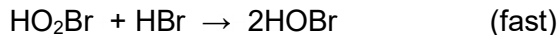


- 11** The reaction between HBr and O₂ is thought to occur via a multi-step mechanism:



The overall reaction is $4\text{HBr} + \text{O}_2 \rightarrow 2\text{Br}_2 + 2\text{H}_2\text{O}$.

Which statement is correct?

- A** The overall order of reaction is 3.
- B** HO₂Br is the only intermediate in the reaction.
- C** HOBr acts as a catalyst in the reaction.
- D** Units of the rate constant is mol⁻¹ dm³ s⁻¹.